OLA RIDE DEMAND & CANCELLATION ANALYSIS

Business Health, Operations & Financial Insights

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Github

★ Unlocking Business Value with Analytics

★ Driving Smarter Decisions Through Data

CONTEXT & PROBLEM STATEMENT

Context:

Ride-hailing platforms like Ola depend on efficient ride completion and customer satisfaction. High cancellations lead to revenue loss and poor user experience.

Core Problem:

Ola is experiencing a high cancellation rate (~33%), affecting revenue and operational efficiency.

Objective:

Identify key drivers of cancellations and improve ride completion, customer experience, and revenue.

DATA ENGINEERING SOURCE TO PIPELINE

Source Dataset:

- Name: Ola Ride Dataset (Kaggle)
- Size: 50,000 rows × 21 columns
- Kaggle: Dataset

Python ETL: Major Cleaning Steps:

- **Handling Missing Values:** Filled null values in booking status, ratings, and ride details to ensure data consistency
- **Data Standardization:** Standardized vehicle types and location names for uniform analysis
- **Removing Inconsistencies:** Removed duplicate and invalid ride records to improve data quality
- **Data Type Conversion:** Converted timestamp fields into datetime format for time-based analysis
- **Feature Engineering:** Created key metrics like revenue, cancellation rate, and completion rate for business analysis
- **Created key metrics:** Revenue → based on ride value
  Cancellation Rate → cancelled rides / total requests
  Completion Rate → completed rides / total requests

KPI & METRICS FRAMEWORK

Business Performance KPIs

- **Total Requests:** 49,866 → Overall demand volume
- **Revenue Earned:** ₹34.18M → Total successful ride revenue
- **Revenue Lost:** ₹16.84M → Financial loss due to cancellations
- **Recovery Potential:** ₹2.52M → Opportunity to recover lost revenue

Customer Experience KPIs

- **Avg Customer Rating:** ~4.0 → Service quality
- **Avg Wait Time:** ~15.5 mins → Customer satisfaction indicator

Operational KPIs

- **Completion Rate:** 66.99% → Ride success efficiency
- **Cancellation Rate:** 33.01% → Core business problem

KEY INSIGHTS

- High Cancellation Rate (~33%) is the biggest revenue risk
- Driver cancellations (~19%) significantly reduce ride completion
- Peak demand occurs during evening & night hours, causing supply gaps
- High-demand locations face ride availability issues
- Driver not moving towards pickup location is a major reason for ride cancellations
- Long wait time negatively impacts customer experience and retention

ADVANCED ANALYSIS

Root Cause Analysis

- Driver cancellations
- Driver inactivity or delay in reaching pickup location
- Demand-supply mismatch

Trend Analysis

- Peak hour demand leads to higher cancellations
- Time-based patterns impact ride success

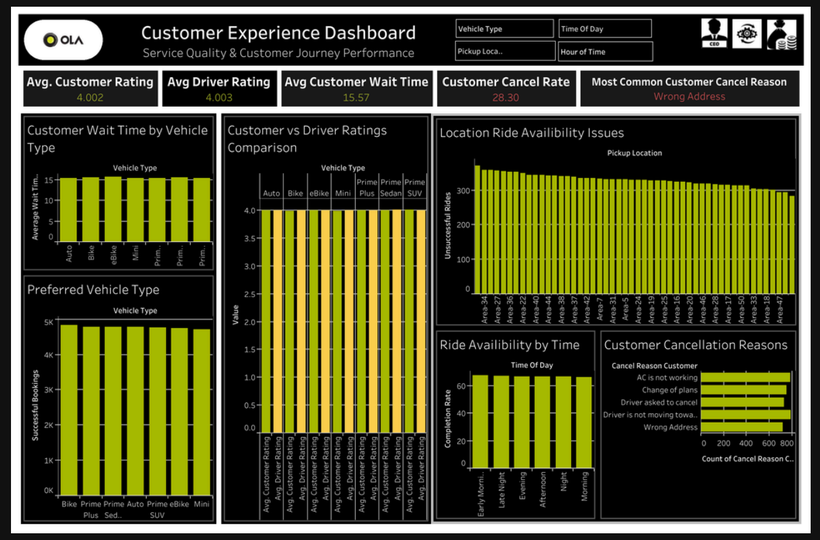
Segmentation Analysis

- Different vehicle types show varying cancellation behavior

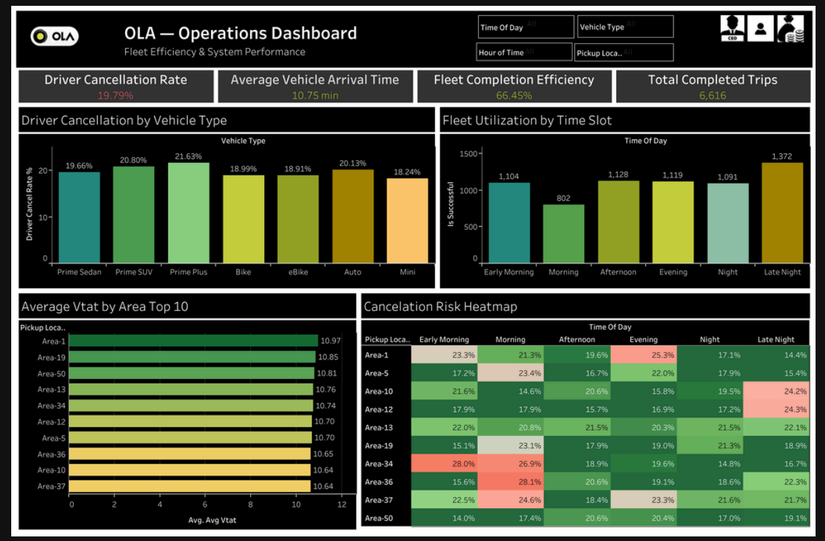
TABLEAU DASHBOARD WALKTHROUGH

Executive Dashboard (CEO View)

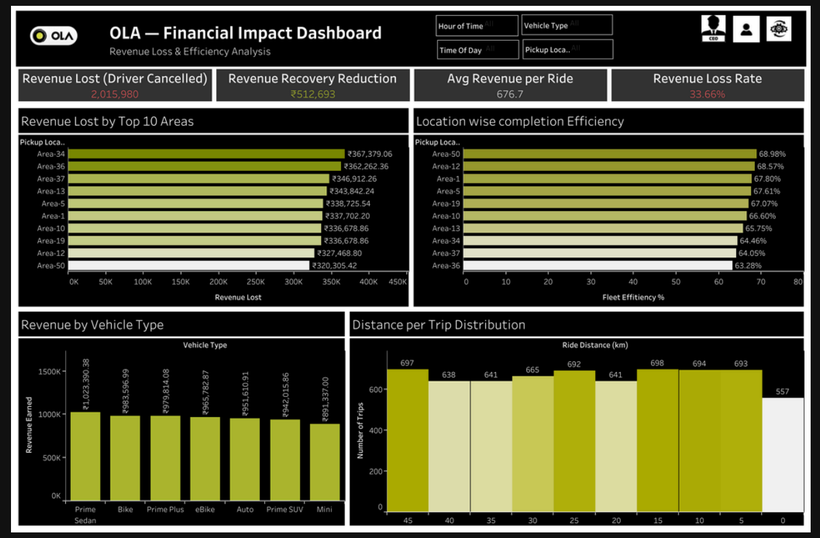
Customer Experience Dashboard



Operations Dashboard



Financial Dashboard



RECOMMENDATIONS

Reduce Driver Cancellations

- Introduce incentives during peak hours
- Improve completion rate and reliability

Ensure Driver Movement

- Monitor driver movement after ride acceptance
- Introduce penalties for unnecessary delays

Manage Peak Hour Demand

- Apply surge pricing and dynamic driver allocation
- Improve ride availability

Reduce Customer Wait Time

- Optimize driver distribution across locations
- Enhance customer experience

IMPACT & VALUE

- Reduce cancellation rate by 10–15%
- Increase revenue by ₹5–10 lakh
- Improve completion rate to 75%+
- Enhance customer satisfaction and retention
- Better demand prediction and operational efficiency

LIMITATIONS & NEXT STEPS

Limitations

- No real-time ride data
- Limited customer behavioral insights
- No external factors (traffic, weather, events)
- Analysis based on historical patterns

Next Steps

- Build ML model for cancellation prediction
- Develop real-time monitoring dashboard
- Implement demand forecasting system
- Optimize driver allocation using data

THANK YOU